# On $C^*$-Algebras and the Notation †: Bounded Operators, Adjoints, and the Dirac Formalism


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**Abstract**

This note clarifies the relation between the standard mathematical notion of a $C^*$-algebra and the notation $C^\dagger$, used in physics. We show that both notions coincide once the involution $*$ is identified with the Hilbert (Hermitian) adjoint †. We then develop, in a self-contained way, the underlying operator-theoretic framework: bounded operators on a Hilbert space, the operator norm, the definition of the adjoint, and the key identities that relate bras, kets, and operators under Hermitian conjugation in Dirac notation.


## 1 Introduction

In the theory of operator algebras, a $C^*$*-algebra* is the standard and universally accepted structure: a complex Banach algebra equipped with an abstract involution satisfying the $C^*$-identity. In the physics literature, particularly in quantum mechanics and quantum information, the same involution is often denoted by the dagger symbol †, since in concrete representations it coincides with the Hermitian adjoint of an operator. This has led to the informal use of the notation "$C^\dagger$-algebra." The purpose of this note is threefold:

(i) State precisely the definition of a $C^*$-algebra and clarify that $C^\dagger$ is not an independent mathematical structure, but simply a notational variant used in physics.

(ii) Develop the operator-theoretic background (bounded operators, operator norm, adjoint) that makes this identification precise.

(iii) Derive, carefully and from first principles, the identities that govern the interaction of the adjoint with bras and kets in Dirac notation.

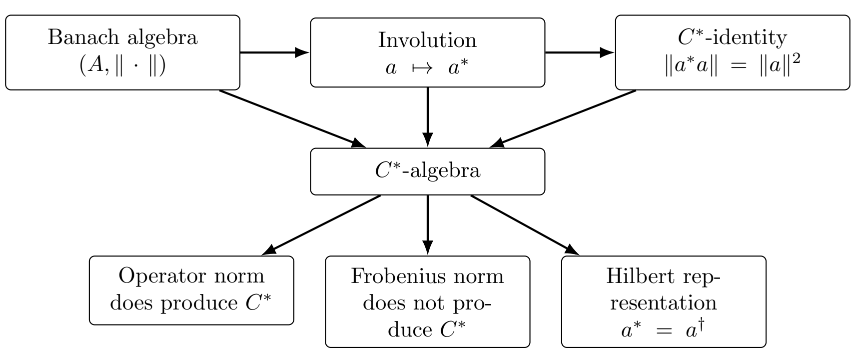

**Figure 1:** Conceptual map: a $C^*$-algebra combines a Banach algebra, an involution, and the $C^*$-identity.

## 2 $C^*$-Algebras and the Notation †

### 2.1 The standard definition

**Definition 1** (Complex Banach algebra)**.** *A* complex Banach algebra *is a complex vector space $A$ equipped with a product*

$$A \times A \longrightarrow A, \qquad (a,b) \longmapsto ab, \tag{1}$$

*and with a norm*

$$\|\cdot\| : A \longrightarrow [0,\infty), \tag{2}$$

*such that, for all $a,b,c \in A$ and $\lambda \in \mathbb{C}$,*

$$(a+b)c = ac + bc, \qquad a(b+c) = ab + ac, \tag{3}$$

$$(\lambda a)b = a(\lambda b) = \lambda(ab), \qquad (ab)c = a(bc), \tag{4}$$

$$\|a\| \geq 0, \qquad \|a\| = 0 \Longleftrightarrow a = 0, \tag{5}$$

$$\|\lambda a\| = |\lambda|\, \|a\|, \qquad \|a+b\| \leq \|a\| + \|b\|, \tag{6}$$

$$\|ab\| \leq \|a\|\, \|b\|. \tag{7}$$

*Moreover, $A$ is* complete *with respect to that norm: if $(a_n)$ is a Cauchy sequence, that is, if for every $\varepsilon > 0$ there exists $N$ such that*

$$\|a_n - a_m\| < \varepsilon \qquad \text{whenever } n, m \geq N, \tag{8}$$

*then there exists $a \in A$ such that*

$$\|a_n - a\| \longrightarrow 0. \tag{9}$$

**Example 1** (The complex numbers)**.** *The simplest example of a complex Banach algebra is $\mathbb{C}$ itself. As a complex vector space, addition and scalar multiplication are the usual ones; as an algebra, the product is the ordinary multiplication of complex numbers*

$$\mathbb{C} \times \mathbb{C} \longrightarrow \mathbb{C}, \qquad (z,w) \longmapsto zw. \tag{10}$$

*The norm is the modulus*

$$\begin{aligned} |\cdot| &: \mathbb{C} \longrightarrow [0,\infty), \\ z = a + ib &\longmapsto |z| := \sqrt{a^2+b^2}. \end{aligned} \tag{11}$$

*Geometrically, we identify $z = a + ib$ with the point $(a,b)$ in the complex plane; then $|z|$ is the Euclidean distance from the origin to that point.*

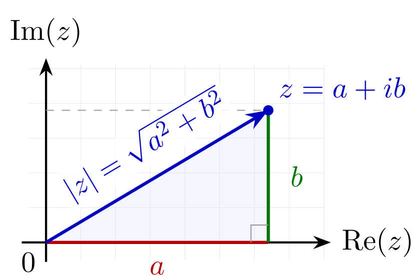


**Figure 2:** The modulus of $z$ is the hypotenuse of the right triangle whose legs are its real and imaginary parts.

*With this norm, for all $z, w \in \mathbb{C}$ and $\lambda \in \mathbb{C}$, one has*

$$
\begin{aligned}
&|z| \geq 0, && |z| = 0 \Longleftrightarrow z = 0, && (12)\\
&|\lambda z| = |\lambda|\,|z|, && |z+w| \leq |z| + |w|, && (13)\\
&|zw| = |z|\,|w| \leq |z|\,|w|. && && (14)
\end{aligned}
$$

*The last equality shows, in particular, the compatibility of the norm with the product. Moreover, $\mathbb{C}$ is complete: if $(z_n)$ is a Cauchy sequence and we write $z_n = a_n + ib_n$, then $(a_n)$ and $(b_n)$ are Cauchy sequences in $\mathbb{R}$; since $\mathbb{R}$ is complete, there exist $a, b \in \mathbb{R}$ such that $a_n \to a$ and $b_n \to b$. Therefore*

$$z_n \longrightarrow a + ib \in \mathbb{C}. \tag{15}$$

*Thus, $\mathbb{C}$ with the usual modulus is a complex Banach algebra.*

**Example 2** ($\mathbb{C}^n$ with coordinatewise product)**.** *Consider $\mathbb{C}^n$ and let $\boldsymbol{z}, \boldsymbol{w} \in \mathbb{C}^n$, with*

$$\boldsymbol{z} = (z_1, \ldots, z_n), \qquad \boldsymbol{w} = (w_1, \ldots, w_n), \tag{16}$$

*We define the usual sum and the coordinatewise product by*

$$\boldsymbol{zw} := (z_1 w_1, \ldots, z_n w_n). \tag{17}$$

*The norm we use is the Euclidean norm*

$$\|\boldsymbol{z}\| := \sqrt{|z_1|^2 + \cdots + |z_n|^2}. \tag{18}$$

*With these operations, $\mathbb{C}^n$ is a normed complex algebra. The compatibility of the norm with the product is verified as follows:*

$$
\begin{aligned}
\|\boldsymbol{zw}\|^2 &= |z_1 w_1|^2 + \cdots + |z_n w_n|^2 && (19)\\
&\leq \left(|z_1|^2 + \cdots + |z_n|^2\right)\left(|w_1|^2 + \cdots + |w_n|^2\right) && (20)\\
&= \|\boldsymbol{z}\|^2 \|\boldsymbol{w}\|^2. && (21)
\end{aligned}
$$

*Therefore,*

$$\|\boldsymbol{zw}\| \leq \|\boldsymbol{z}\|\|\boldsymbol{w}\|. \tag{22}$$

*Finally, since $\mathbb{C}^n$ is finite-dimensional and each coordinate lives in $\mathbb{C}$, every Cauchy sequence converges coordinate by coordinate in $\mathbb{C}^n$. Thus, $\mathbb{C}^n$ with coordinatewise product and the Euclidean norm is a complex Banach algebra.*

**Example 3** (Complex matrices with the operator norm)**.** *For each $n \geq 1$, the space of complex matrices*

$$M_n(\mathbb{C}) \tag{23}$$

*is a complex algebra with the usual matrix sum and product. Using on $\mathbb{C}^n$ the Euclidean norm defined above, for $A \in M_n(\mathbb{C})$ we define its* operator norm[1] *by*

$$\boxed{\|A\|_{\text{op}} := \sup\left\{\|Av\| : \ v \in \mathbb{C}^n, \ \|v\| = 1\right\}.} \tag{24}$$

[1] Historically, this way of measuring operators became consolidated with the development of functional analysis in the 1920s and 1930s. In particular, the Banach–Steinhaus uniform boundedness principle (1927) formulated the importance of controlling families of operators through their norms, and Stefan Banach's book *Théorie des opérations linéaires* (1932) systematized the study of bounded linear operators between complete normed spaces.

Thus, rather than being the point invention of a single person, the operator norm is part of the very birth of the modern theory of Banach spaces and linear operators.

*That is, we consider all unit vectors $v$ in $\mathbb{C}^n$, compute the Euclidean length of $Av$, and take the supremum of those real numbers. Thus, $\|A\|_{\mathrm{op}}$ measures the maximum factor by which $A$ can stretch a unit vector.*

**Remark 1** (Why this supremum exists for matrices)**.** *In this example we do not need to assume anything additional about $A$: every matrix $A \in M_n(\mathbb{C})$ defines a linear operator on the finite-dimensional space $\mathbb{C}^n$. Moreover, the unit sphere*

$$S^{2n-1} := \{v \in \mathbb{C}^n : \|v\| = 1\} \tag{25}$$

*is compact, and the function*

$$S^{2n-1} \longrightarrow [0, \infty), \qquad v \longmapsto \|Av\| \tag{26}$$

*is continuous. By the extreme value theorem,*[2] *the supremum above exists and is in fact attained: there is a unit vector $v_0$ such that*

$$\|A\|_{\mathrm{op}} = \|Av_0\|. \tag{27}$$

*For this reason, in finite dimension, we could also write* max *instead of* sup.

*Let us verify that with this norm $M_n(\mathbb{C})$ is a Banach algebra. We do this in parts.*

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[2] Extreme value theorem: if $K$ is compact and $f : K \to \mathbb{R}$ is continuous, then there exist $x_{\min}, x_{\max} \in K$ such that

$$f(x_{\min}) \le f(x) \le f(x_{\max}) \qquad \text{for all } x \in K. \tag{I}$$

Proof. First, $f(K)$ is compact: if $\{U_\alpha\}$ is an open cover of $f(K)$, then $\{f^{-1}(U_\alpha)\}$ is an open cover of $K$; by compactness of $K$, there exists a finite subcover

$$f^{-1}(U_{\alpha_1}), \dots, f^{-1}(U_{\alpha_m}), \tag{II}$$

and then $U_{\alpha_1}, \dots, U_{\alpha_m}$ cover $f(K)$. Thus, $f(K)$ is compact in $\mathbb{R}$. In $\mathbb{R}$, every compact set is closed and bounded.

We now use the supremum axiom: every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum. Since $f(K)$ is nonempty and bounded above, there exists

$$M := \sup f(K). \tag{III}$$

To prove that $M \in f(K)$, suppose the contrary. Since $M$ is the supremum, $M$ is an upper bound for $f(K)$ and no quantity smaller than $M$ can be an upper bound. Therefore, for each $n \ge 1$, the number $M - \frac{1}{n}$ is not an upper bound for $f(K)$; hence there exists $y_n \in f(K)$ such that

$$M - \frac{1}{n} < y_n. \tag{IV}$$

Moreover, since $M$ is an upper bound, one has $y_n \le M$. Thus,

$$M - \frac{1}{n} < y_n \le M, \tag{V}$$

and therefore $y_n \to M$. Since $f(K)$ is closed, it contains the limit of every convergent sequence of points of $f(K)$, so $M \in f(K)$, contradicting the initial supposition.

Therefore there exists $x_{\max} \in K$ such that

$$f(x_{\max}) = M. \tag{VI}$$

The minimum is obtained in the same way using $m := \inf f(K)$, or by applying the previous argument to $-f$. In our case,

$$K = S^{2n-1} \qquad \text{y} \qquad f(v) = \|Av\|. \tag{VII}$$

□

(a) ***Basic norm properties.*** *For all* $A, B \in M_n(\mathbb{C})$ *and* $\lambda \in \mathbb{C}$,

$$\|A\|_{\mathrm{op}} \geq 0, \tag{28}$$

$$\|\lambda A\|_{\mathrm{op}} = |\lambda|\,\|A\|_{\mathrm{op}}, \tag{29}$$

$$\|A + B\|_{\mathrm{op}} \leq \|A\|_{\mathrm{op}} + \|B\|_{\mathrm{op}}. \tag{30}$$

*The triangle inequality is obtained because, for every unit vector* $v$,

$$\|(A + B)v\| \leq \|Av\| + \|Bv\| \leq \|A\|_{\mathrm{op}} + \|B\|_{\mathrm{op}}. \tag{31}$$

*Taking the supremum over all unit vectors* $v$*, the left-hand side becomes* $\|A + B\|_{\mathrm{op}}$*, and we obtain precisely (30). Moreover, if* $\|A\|_{\mathrm{op}} = 0$*, then* $\|Av\| = 0$ *for every unit vector* $v$*; by rescaling,* $Aw = 0$ *for every* $w \in \mathbb{C}^n$*, and therefore* $A = 0$.

(b) ***Compatibility with the product.*** *First note that the definition of* $\|A\|_{\mathrm{op}}$*, although taken over unit vectors, implies that for every* $w \in \mathbb{C}^n$ *one has*

$$\boxed{\|Aw\| \leq \|A\|_{\mathrm{op}}\|w\|.} \tag{32}$$

*Indeed, if* $w = 0$ *the inequality is immediate; if* $w \neq 0$*, then* $w/\|w\|$ *is unitary, and therefore*

$$\left\| A\left(\frac{w}{\|w\|}\right) \right\| \leq \|A\|_{\mathrm{op}}. \tag{33}$$

*Multiplying by* $\|w\|$*, we obtain the previous inequality.*

*Now let* $v$ *be a unit vector, that is,* $\|v\| = 1$*. Since* $ABv = A(Bv)$*, we apply the previous inequality with* $w := Bv$:

$$\|ABv\| = \|A(Bv)\| \leq \|A\|_{\mathrm{op}}\|Bv\| \leq \|A\|_{\mathrm{op}}\|B\|_{\mathrm{op}}. \tag{34}$$

*Taking the supremum over all unit vectors* $v$*, we obtain submultiplicativity:*

$$\|AB\|_{\mathrm{op}} \leq \|A\|_{\mathrm{op}}\|B\|_{\mathrm{op}}, \tag{35}$$

(c) ***Completeness.*** *Let* $(A_k)$ *be a Cauchy sequence in the operator norm. If* $e_1, \ldots, e_n$ *is the canonical basis of* $\mathbb{C}^n$*, recall that multiplying a matrix by* $e_j$ *extracts its* $j$*-th column. For example, in dimension* 3,

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a_{12} \\ a_{22} \\ a_{32} \end{pmatrix}. \tag{36}$$

*Thus,* $Ae_j$ *is precisely the* $j$*-th column of* $A$*. Then*

$$\|(A_k - A_\ell)e_j\| \leq \|A_k - A_\ell\|_{\mathrm{op}}, \qquad j = 1, \ldots, n. \tag{37}$$

*This inequality says that closeness of matrices in operator norm controls closeness of each column* $j$*. Indeed, given* $\varepsilon > 0$*, since* $(A_k)$ *is Cauchy, there exists* $N$ *such that*

$$\|A_k - A_\ell\|_{\mathrm{op}} < \varepsilon \qquad \text{if } k, \ell \geq N. \tag{38}$$

*By the previous inequality, for each $j = 1, \dots, n$ one then has*

$$\|A_k e_j - A_\ell e_j\| = \|(A_k - A_\ell)e_j\| < \varepsilon, \qquad k, \ell \geq N. \tag{39}$$

*Thus, for each $j$, the sequence of columns $A_k e_j$ is Cauchy in $\mathbb{C}^n$. Since $\mathbb{C}^n$ is complete with the Euclidean norm, there exists a vector $a_j \in \mathbb{C}^n$ such that*

$$A_k e_j \longrightarrow a_j. \tag{40}$$

*We define $A \in M_n(\mathbb{C})$ to be the matrix whose columns are $a_1, \dots, a_n$. Now take a unit vector*

$$v = \sum_{j=1}^n v_j e_j. \tag{41}$$

*By linearity of $A_k - A$,*

$$(A_k - A)v = (A_k - A)\left(\sum_{j=1}^n v_j e_j\right) = \sum_{j=1}^n v_j (A_k - A)e_j. \tag{42}$$

*Applying the triangle inequality to this finite sum, we obtain*

$$\|(A_k - A)v\| \leq \sum_{j=1}^n |v_j| \, \|(A_k - A)e_j\|. \tag{43}$$

*Since $v$ is unitary, $\sum_{j=1}^n |v_j|^2 = 1$. By Cauchy–Schwarz,*[3]

$$\sum_{j=1}^n |v_j| \, \|(A_k - A)e_j\| \leq \left(\sum_{j=1}^n \|(A_k - A)e_j\|^2\right)^{1/2}. \tag{44}$$

*For each $j$, the factor*

$$\|(A_k - A)e_j\| = \|A_k e_j - A e_j\| \tag{45}$$

*tends to $0$, because $A_k e_j \to a_j$ and $Ae_j = a_j$ by definition of $A$. Since there are only finitely many columns, the right-hand side also tends to $0$. Taking the supremum over all unit vectors $v$, we obtain*

$$\|A_k - A\|_{\mathrm{op}} \longrightarrow 0, \tag{46}$$

*and every Cauchy sequence converges in $M_n(\mathbb{C})$.*

□

The previous example allowed us to observe concretely what convergence in the operator norm means for matrices: a Cauchy sequence of matrices converges to the matrix constructed column by column. Indeed, if the columns $A_k e_j$ converge to $a_j$, then the limit is precisely the matrix $A$ whose columns are $a_1, \dots, a_n$. This is the concrete reason why $M_n(\mathbb{C})$ is complete with this norm.

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[3] The Cauchy–Schwarz inequality says that, for complex numbers $a_j, b_j$,

$$\left|\sum_{j=1}^n a_j \overline{b_j}\right| \leq \left(\sum_{j=1}^n |a_j|^2\right)^{1/2} \left(\sum_{j=1}^n |b_j|^2\right)^{1/2}. \tag{VIII}$$

Here it is applied with $a_j = |v_j|$ and $b_j = \|(A_k - A)e_j\|$.

**Example 4** (Complex matrices with the Frobenius norm)**.** *On the same space $M_n(\mathbb{C})$ one can also define the* Frobenius norm *or* Hilbert–Schmidt norm.[4] *If $A = (a_{ij})$, we set*

$$\|A\|_{\mathrm{Frob}} := \left( \sum_{i,j=1}^{n} |a_{ij}|^2 \right)^{1/2} . \tag{47}$$

*This norm measures the matrix as if it were a vector with $n^2$ entries. To see this formally, we identify*

$$A = (a_{ij}) \longleftrightarrow (a_{11}, a_{12}, \ldots, a_{1n}, a_{21}, \ldots, a_{nn}) \in \mathbb{C}^{n^2}. \tag{48}$$

*Under this identification, $\|A\|_{\mathrm{Frob}}$ is the usual Euclidean norm of that vector of entries. Therefore it satisfies, for all $A, B \in M_n(\mathbb{C})$ and $\lambda \in \mathbb{C}$,*

$$\|A\|_{\mathrm{Frob}} \geq 0, \tag{49}$$
$$\|A\|_{\mathrm{Frob}} = 0 \iff A = 0, \tag{50}$$
$$\|\lambda A\|_{\mathrm{Frob}} = |\lambda| \, \|A\|_{\mathrm{Frob}}, \tag{51}$$
$$\|A + B\|_{\mathrm{Frob}} \leq \|A\|_{\mathrm{Frob}} + \|B\|_{\mathrm{Frob}}. \tag{52}$$

*Moreover, just as in any finite-dimensional vector space, every Cauchy sequence of matrices converges entry by entry to another matrix in $M_n(\mathbb{C})$. Thus, $M_n(\mathbb{C})$ is also a Banach space with the Frobenius norm.*

**Definition 2** ($C^*$-algebra)**.** *A* $C^*$-algebra *is a complex Banach algebra $A$ equipped with an involution*

$$* : A \longrightarrow A, \qquad a \mapsto a^*, \tag{53}$$

*which satisfies, for all $a, b \in A$ and $\lambda, \mu \in \mathbb{C}$,*

$$(ab)^* = b^* a^*, \tag{54}$$
$$(a^*)^* = a, \tag{55}$$
$$(\lambda a + \mu b)^* = \overline{\lambda}\, a^* + \overline{\mu}\, b^*, \tag{56}$$

*together with the $C^*$*-identity

$$\boxed{\|a^* a\| = \|a\|^2.} \tag{57}$$

**Example 5** (The standard matrix $C^*$-algebra)**.** *In the standard matrix case, this definition is realized by taking on $M_n(\mathbb{C})$ the involution given by the conjugate transpose*

$$\boxed{A^* := \overline{A}^{T}.} \tag{58}$$

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[4] In finite dimension these two names refer to the same norm. The Hilbert–Schmidt inner product on $M_n(\mathbb{C})$ is defined by

$$\langle A, B \rangle_{\mathrm{HS}} := \mathrm{Tr}(A^* B) \tag{IX}$$

and the induced norm is

$$\|A\|_{\mathrm{HS}} := \sqrt{\langle A, A \rangle_{\mathrm{HS}}} = \sqrt{\mathrm{Tr}(A^* A)} = \left( \sum_{i,j=1}^{n} |a_{ij}|^2 \right)^{1/2} = \|A\|_{\mathrm{Frob}}. \tag{X}$$

The name "Frobenius" is common in matrix linear algebra; "Hilbert–Schmidt" is used more often when matrices are thought of as operators between Hilbert spaces.

*With the operator norm, the $C^*$-identity then holds:*

$$\boxed{\|A^*A\|_{\mathrm{op}} = \|A\|_{\mathrm{op}}^2.} \tag{59}$$

*To prove this, take $H := A^*A$. This matrix is positive Hermitian: this means that $H^* = H$ and that*

$$\langle Hv, v\rangle \geq 0 \qquad \text{for all } v \in \mathbb{C}^n. \tag{60}$$

*In particular, its eigenvalues $\lambda_1, \ldots, \lambda_n$ are nonnegative real numbers. Let us first prove that, for a positive Hermitian matrix $H$, its operator norm is the largest of its eigenvalues. By the spectral theorem[5] there exists an orthonormal basis of eigenvectors $e_1, \ldots, e_n$ such that $He_j = \lambda_j e_j$. If $v$ is unitary and*

$$v = \sum_{j=1}^{n} c_j e_j, \qquad \sum_{j=1}^{n} |c_j|^2 = 1, \tag{61}$$

*then*

$$Hv = \sum_{j=1}^{n} \lambda_j c_j e_j, \tag{62}$$

*and by orthonormality*

$$\|Hv\|^2 = \sum_{j=1}^{n} \lambda_j^2 |c_j|^2 \leq \left(\max_j \lambda_j\right)^2 \sum_{j=1}^{n} |c_j|^2 = \left(\max_j \lambda_j\right)^2. \tag{63}$$

*Therefore, for every unit vector $v$,*

$$\|Hv\| \leq \max_{1\leq j\leq n} \lambda_j. \tag{64}$$

*Moreover, since the maximum is attained among the eigenvalues, there exists a unit eigenvector $e_{\max}$ associated with $\max_j \lambda_j$. Then*

$$\|He_{\max}\| = \left\| \left(\max_j \lambda_j\right) e_{\max} \right\| = \max_j \lambda_j. \tag{65}$$

*Since the previous inequality holds for every unit vector $v$, taking the supremum gives*

$$\|H\|_{\mathrm{op}} \leq \max_j \lambda_j. \tag{66}$$

*On the other hand, evaluating at the unit vector $e_{\max}$,*

$$\max_j \lambda_j = \|He_{\max}\| \leq \|H\|_{\mathrm{op}} \leq \max_j \lambda_j. \tag{67}$$

*That is, we have reached the following theorem.*

**Theorem 1** (Norm of a positive Hermitian matrix)**.** *If $H \in M_n(\mathbb{C})$ is a positive Hermitian matrix with eigenvalues $\lambda_1, \ldots, \lambda_n$, then*

$$\boxed{\|H\|_{\mathrm{op}} = \max_j \lambda_j.} \tag{68}$$

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[5] In finite dimension, the spectral theorem states that every Hermitian matrix can be diagonalized by an orthonormal basis of eigenvectors. That is, there exist orthonormal vectors $e_1, \ldots, e_n$ and real scalars $\lambda_1, \ldots, \lambda_n$ such that $He_j = \lambda_j e_j$.

*Since $H = A^*A$, we conclude that*

$$\boxed{\|A^*A\|_{\mathrm{op}} = \max_j \lambda_j.} \tag{69}$$

*Now we use $H = A^*A$ to compute the norm of $A$ directly. If $v$ is unitary and*

$$v = \sum_{j=1}^{n} c_j e_j, \qquad \sum_{j=1}^{n} |c_j|^2 = 1, \tag{70}$$

*then*

$$\|Av\|^2 = \langle Av, Av\rangle = \langle A^*Av, v\rangle = \langle Hv, v\rangle. \tag{71}$$

*Since $He_j = \lambda_j e_j$, we obtain*

$$\langle Hv, v\rangle = \sum_{j=1}^{n} \lambda_j |c_j|^2 \le \left(\max_j \lambda_j\right) \sum_{j=1}^{n} |c_j|^2 = \max_j \lambda_j. \tag{72}$$

*Since this holds for every unit vector $v$, taking the supremum gives*

$$\|A\|_{\mathrm{op}}^2 = \sup_{\|v\|=1} \|Av\|^2 \le \max_j \lambda_j. \tag{73}$$

*On the other hand, taking $v = e_{\max}$,*

$$\max_j \lambda_j = \langle He_{\max}, e_{\max}\rangle = \|Ae_{\max}\|^2 \le \|A\|_{\mathrm{op}}^2. \tag{74}$$

*Therefore,*

$$\boxed{\|A\|_{\mathrm{op}}^2 = \sup_{\|v\|=1} \|Av\|^2 = \max_j \lambda_j = \|A^*A\|_{\mathrm{op}}.} \tag{75}$$

*This proves the $C^*$-identity for the operator norm in the standard matrix case.* □

**Example 6** (The Frobenius norm does not satisfy the $C^*$-identity)**.** *The Frobenius norm is a natural norm on $M_n(\mathbb{C})$, but with the standard matrix involution*

$$A^* := \overline{A}^T \tag{76}$$

*it does not satisfy the $C^*$-identity in general. To see this explicitly, take*

$$\sigma_x := \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{77}$$

*For the Frobenius norm,*

$$\|\sigma_x\|_{\mathrm{Frob}} = \left(|0|^2 + |1|^2 + |1|^2 + |0|^2\right)^{1/2} = \sqrt{2}, \tag{78}$$

*whereas*

$$\|\sigma_x^*\sigma_x\|_{\mathrm{Frob}} = \|I_2\|_{\mathrm{Frob}} = \sqrt{2} \ne 2 = \|\sigma_x\|_{\mathrm{Frob}}^2. \tag{79}$$

*Thus, with the Frobenius norm, $\|\sigma_x^*\sigma_x\|_{\mathrm{Frob}} \ne \|\sigma_x\|_{\mathrm{Frob}}^2$. The difference is not in the involution, but in the norm.*

The two previous examples show the contrast we need: the operator norm does respect the $C^*$-identity, whereas the Frobenius norm, although perfectly natural as a matrix norm, fails precisely at that compatibility. The following result explains that this is not a coincidence: for the standard matrix involution, the $C^*$-identity forces the operator norm.

**Theorem 2** (The $C^*$-identity forces the operator norm)**.** *Let $M_n(\mathbb{C})$ be equipped with the standard matrix involution*

$$A^* := \overline{A}^T. \tag{80}$$

*If $\|\cdot\|$ is a norm that makes $M_n(\mathbb{C})$ into a $C^*$-algebra, then*

$$\boxed{\|A\| = \|A\|_{\mathrm{op}} \qquad \text{for all } A \in M_n(\mathbb{C}).} \tag{81}$$

*In other words, the operator norm is the unique norm compatible with the standard matrix $C^*$-structure.*

*Proof.* The $C^*$-identity leaves no freedom in choosing the norm. Indeed, for any matrix $A$ one has

$$\boxed{\|A\|^2 = \|A^*A\|.} \tag{82}$$

Thus, the norm of $A$ is determined by the matrix $A^*A$. The point that remains to justify is how to read the norm of that positive matrix. For this we use the following auxiliary claim.

**Auxiliary claim.** If $B \in M_n(\mathbb{C})$ is normal, that is, $B^*B = BB^*$, then

$$\boxed{\|B\| = r(B),} \tag{83}$$

where

$$r(B) := \max\{|\lambda| : \lambda \in \sigma(B)\} \tag{84}$$

is the spectral radius of $B$, and

$$\sigma(B) := \{\lambda \in \mathbb{C} : \lambda I - B \text{ is not invertible}\} \tag{85}$$

is the spectrum of $B$. In the matrix case this definition is completely algebraic, because

$$\lambda I - B \text{ is not invertible} \quad \Longleftrightarrow \quad \det(\lambda I - B) = 0. \tag{86}$$

Here a matrix is *invertible* if there exists another matrix that multiplies it on both sides and gives the identity, and det denotes the determinant. The roots of $\det(\lambda I - B)$ are precisely the eigenvalues of $B$. Therefore,

$$r(B) = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } B\}. \tag{87}$$

This is the important subtlety: below we will use the spectral radius formula, which does write $r(B)$ in terms of the unknown $C^*$-norm we are trying to identify; however, the number $r(B)$ is already fixed by the matrix itself, since it depends only on its eigenvalues. In particular, if $B$ is positive Hermitian, then its eigenvalues are nonnegative real numbers and

$$\boxed{\|B\| = \max_j \lambda_j(B).} \tag{88}$$

*Proof of the claim.* The fundamental result that allows one to pass from the spectrum to the norm is the following theorem of Gelfand[6].

**Gelfand result (spectral radius formula).** If $A$ is a unital Banach algebra, that is, a Banach algebra with an identity element $1_A$ for the product, and $a \in A$, then

$$\boxed{r(a) = \lim_{m\to\infty} \|a^m\|^{1/m}.} \tag{89}$$

This result is deep because it identifies two objects that, at first sight, live on different levels. The left-hand side, $r(a)$, is defined by the spectrum:

$$r(a) = \sup\{|\lambda| : \lambda \in \sigma(a)\}. \tag{90}$$

The right-hand side, by contrast, measures the asymptotic growth of the powers of $a$ using the norm of the algebra.

In our argument this is exactly what we need: although we are working with an unknown $C^*$-norm, the spectral radius remains the spectral radius of the element. Applying Gelfand's formula to $B$ we obtain

$$r(B) = \lim_{m\to\infty} \|B^m\|^{1/m}. \tag{91}$$

The equality $r(B) = \|B\|$ does not come from Gelfand alone; it appears now by using the fact that $B$ is normal and that the norm satisfies the $C^*$-identity. Since $B$ is normal, one has

$$\|B^2\|^2 = \|(B^2)^* B^2\| = \|(B^*B)^2\| = \|(B^*B)^*(B^*B)\| = \|B^*B\|^2 = \|B\|^4. \tag{92}$$

Therefore, $\|B^2\| = \|B\|^2$. Since every power of a normal element is again normal, repeating the argument we obtain

$$\|B^{2^k}\| = \|B\|^{2^k}, \qquad k = 1, 2, \ldots \tag{93}$$

Taking the subsequence $m = 2^k$ in the spectral radius formula,

$$r(B) = \lim_{k\to\infty} \|B^{2^k}\|^{1/2^k} = \|B\|. \tag{94}$$

If, moreover, $B$ is positive Hermitian, its eigenvalues are nonnegative real numbers; hence $r(B) = \max_j \lambda_j(B)$. This proves the claim. □

Now we apply the claim to $B = A^*A$. Since $A^*A$ is positive Hermitian,

$$\|A\|^2 = \|A^*A\| = \max_j \lambda_j(A^*A). \tag{95}$$

By the calculation made in the standard matrix example,

$$\|A\|_{\mathrm{op}}^2 = \max_j \lambda_j(A^*A). \tag{96}$$

Therefore,

$$\|A\| = \|A\|_{\mathrm{op}}, \tag{97}$$

as we wanted to prove. □

[6]Israel M. Gelfand (1913-2009) was one of the central mathematicians of the twentieth century. His work was decisive in functional analysis, representation theory, distributions, and normed algebras; in particular, Gelfand theory allows one to study algebras through their spectrum, which is exactly the idea we are using here.

**Remark 2** (Why the operator norm)**.** *The previous theorem and the Frobenius example show the essential point. The Frobenius norm does make $M_n(\mathbb{C})$ into a Banach algebra, but it is not compatible with the involution in the sense of the $C^*$-identity. The operator norm, by contrast, does satisfy*

$$\boxed{\|A^*A\|_{\mathrm{op}} = \|A\|_{\mathrm{op}}^2,} \tag{98}$$

*and in fact it is the unique norm that makes the complex matrices, with conjugate transpose, into the standard matrix $C^*$-algebra.*

*The conceptual difference is the following. For a column vector $x \in \mathbb{C}^n$, the expression*

$$\overline{x}^T x \tag{99}$$

*is a positive scalar and coincides with $\|x\|^2$. By contrast, for a matrix $A \in M_n(\mathbb{C})$,*

$$A^*A \tag{100}$$

*is no longer a scalar, but another matrix. Therefore the equality $\|A^*A\| = \|A\|^2$ does not belong to the definition of a Banach algebra: it is the additional axiom that distinguishes a $C^*$-algebra. This observation is fundamental for quantum mechanics: observables and operations on states are modeled by operators, and the operator norm is the one that measures their action on state vectors in a way compatible with the adjoint.*

The map $*$ is called the *adjoint* or *involution*.

**Example 7** (The commutative algebra $C(X)$)**.** *For the commutative $C^*$-algebra $C(X)$ of continuous complex-valued functions on a compact space*[7] *$X$, the involution is*

$$f^*(x) := \overline{f(x)}, \tag{101}$$

*which is complex conjugation.*

## 2.2 The physical notation $C^\dagger$

In physics (especially in quantum mechanics and quantum information), the symbol $\dagger$ is used to denote the Hermitian adjoint of an operator on a Hilbert space. Consequently, physicists often write $A^\dagger$ instead of $A^*$, and speak informally of a "$C^\dagger$-algebra." Strictly speaking, $C^\dagger$-algebra is not a standard term in the mathematical literature; the universally accepted notion remains the $C^*$-algebra. Nevertheless, it can be defined as a matter of notation:

**Definition 3** ($C^\dagger$-algebra)**.** *A* $C^\dagger$-algebra *is a complex Banach algebra $A$ equipped with a map*

$$\boxed{\dagger : A \longrightarrow A, \qquad a \mapsto a^\dagger,} \tag{102}$$

*which satisfies, for all $a, b \in A$ and $\lambda, \mu \in \mathbb{C}$,*

$$(ab)^\dagger = b^\dagger a^\dagger, \tag{103}$$

$$(a^\dagger)^\dagger = a, \tag{104}$$

$$(\lambda a + \mu b)^\dagger = \overline{\lambda}\, a^\dagger + \overline{\mu}\, b^\dagger, \tag{105}$$

*together with the* $C^\dagger$-identity

$$\boxed{\|a^\dagger a\| = \|a\|^2.} \tag{106}$$

[7] A topological space $X$ is compact if every open cover of $X$ admits a finite subcover. In metric spaces, one may think of it as an abstract form of "closed and bounded."

# 3 Hilbert Spaces

**Definition 4** (Hilbert space)**.** *A* Hilbert space $\mathcal{H}$ *is a complex vector space equipped with an inner product*

$$\langle \cdot, \cdot \rangle : \mathcal{H} \times \mathcal{H} \longrightarrow \mathbb{C}. \tag{107}$$

*With our convention, this inner product is conjugate-linear in the first variable:*

$$\langle \alpha x + \beta y, z \rangle = \overline{\alpha}\, \langle x, z \rangle + \overline{\beta}\, \langle y, z \rangle, \tag{108}$$

*and linear in the second:*

$$\langle x, \alpha y + \beta z \rangle = \alpha \langle x, y \rangle + \beta \langle x, z \rangle, \qquad \alpha, \beta \in \mathbb{C}. \tag{109}$$

*Here the bar denotes complex conjugation of scalars. Moreover, the inner product satisfies Hermitian symmetry*

$$\langle x, y \rangle = \overline{\langle y, x \rangle}, \tag{110}$$

*and positive definiteness:*

$$\langle x, x \rangle \geq 0, \qquad \langle x, x \rangle = 0 \Longleftrightarrow x = 0. \tag{111}$$

*The norm induced by the inner product is*

$$\|x\| := \sqrt{\langle x, x \rangle}. \tag{112}$$

*Finally, $\mathcal{H}$ is* complete *with respect to this norm: every Cauchy sequence in $\mathcal{H}$ converges to a vector of $\mathcal{H}$.*

**Definition 5** (Linear operator)**.** *If $\mathcal{H}$ is a Hilbert space, a* linear operator *on $\mathcal{H}$ is a function*

$$T : \mathcal{H} \longrightarrow \mathcal{H} \tag{113}$$

*that respects addition and multiplication by scalars:*

$$T(\lambda\psi + \mu\phi) = \lambda T\psi + \mu T\phi, \qquad \forall\, \psi, \phi \in \mathcal{H}, \quad \lambda, \mu \in \mathbb{C}. \tag{114}$$

## 3.1 Identification with the Hilbert adjoint

**Definition 6** (Dirac notation)**.** *In Dirac notation, a vector $\psi \in \mathcal{H}$ is written as a* ket $|\psi\rangle$. *To each ket there corresponds the* bra $\langle\psi|$, *defined through the inner product of $\mathcal{H}$ as the linear functional*[8]

$$\langle\psi| : \mathcal{H} \longrightarrow \mathbb{C}, \qquad |\phi\rangle \longmapsto \langle\psi|\phi\rangle := \langle\psi, \phi\rangle. \tag{115}$$

*If $T : \mathcal{H} \to \mathcal{H}$ is a linear operator, we write $T|\psi\rangle$ for its action on the ket $|\psi\rangle$. Moreover, from a bra $\langle\psi| : \mathcal{H} \to \mathbb{C}$ we can construct a new bra by precomposition with $T$:*

$$\langle\psi|T := \langle\psi| \circ T : \mathcal{H} \longrightarrow \mathbb{C}. \tag{116}$$

*The corresponding diagram is*

[8] A linear functional on $\mathcal{H}$ is a linear map from $\mathcal{H}$ to the field of scalars, in this case $\mathbb{C}$.

$$
\begin{tikzcd}
\mathcal{H} \arrow[r, "T"] \arrow[rr, bend right, "\langle\psi|T := \langle\psi| \circ T"'] & \mathcal{H} \arrow[r, "\langle\psi|"] & \mathbb{C}
\end{tikzcd}
$$

*Therefore, for any ket $|\phi\rangle$, the expression*

$$\boxed{\langle\psi|T|\phi\rangle = (\langle\psi|T)(|\phi\rangle) = \langle\psi|(T|\phi\rangle)} \tag{117}$$

*can be read in two equivalent ways: as the new bra $\langle\psi|T$ evaluated at $|\phi\rangle$, or as the bra $\langle\psi|$ evaluated at the ket $T|\phi\rangle$.*

**Example 8** (States of a qubit)**.** *In $\mathcal{H} = \mathbb{C}^2$, a pure state of a qubit is written*

$$|\psi\rangle := \alpha|0\rangle + \beta|1\rangle, \qquad \alpha, \beta \in \mathbb{C}, \qquad |\alpha|^2 + |\beta|^2 = 1. \tag{118}$$

*Its corresponding bra is*

$$\langle\psi| := \overline{\alpha}\langle 0| + \overline{\beta}\langle 1|. \tag{119}$$

*This example fixes the basic notation that will be used below: a ket represents the state, and its associated bra allows one to form expressions of the form $\langle\psi|T|\psi\rangle$ when $T$ is an operator on $\mathcal{H}$.*

**Definition 7** (Bounded operators and $B(\mathcal{H})$)**.** *Let $\mathcal{H}$ be a Hilbert space. A linear operator $T : \mathcal{H} \to \mathcal{H}$ is said to be* bounded *if there exists a constant $M \geq 0$ such that*

$$\|T|\psi\rangle\| \leq M\||\psi\rangle\|, \qquad \forall\, |\psi\rangle \in \mathcal{H}. \tag{120}$$

*We denote by $B(\mathcal{H})$ the set of all bounded linear operators on $\mathcal{H}$.*

**Definition 8** (Hilbert adjoint)**.** *If $T \in B(\mathcal{H})$, the* Hilbert adjoint *of $T$ is the unique bounded operator $T^\dagger \in B(\mathcal{H})$ characterized by the identity*

$$\boxed{\langle y|T|x\rangle = \langle T^\dagger y|x\rangle, \qquad \forall\, |x\rangle, |y\rangle \in \mathcal{H}.} \tag{121}$$

*Equivalently, taking complex conjugates on both sides, the same condition can be written as*

$$\overline{\langle y|T|x\rangle} = \langle x|T^\dagger|y\rangle, \qquad \forall\, |x\rangle, |y\rangle \in \mathcal{H}. \tag{122}$$

*Here the bar denotes the complex conjugate of the scalar $\langle y|T|x\rangle$.*

**Proposition 1** (Construction of the adjoint)**.** *The operator $T^\dagger$ in the previous definition is constructed by means of the Riesz representation theorem. Indeed, fixing $|y\rangle \in \mathcal{H}$, we consider the functional*

$$F_y : \mathcal{H} \longrightarrow \mathbb{C}, \qquad F_y(|x\rangle) := \langle y|T|x\rangle. \tag{123}$$

*Since $T$ is bounded, there exists a constant $M_T \geq 0$ such that $\|T|x\rangle\| \leq M_T\||x\rangle\|$ for every $|x\rangle \in \mathcal{H}$. Therefore, $F_y$ is linear in $|x\rangle$ and bounded, since*

$$|F_y(|x\rangle)| = |\langle y, Tx\rangle| \leq \|T|x\rangle\|\, \||y\rangle\| \leq \big(M_T\||y\rangle\|\big)\, \||x\rangle\|. \tag{124}$$

*Since $|y\rangle$ is fixed, $M_T\||y\rangle\|$ is a constant; this shows that $F_y$ is a bounded functional of the variable $|x\rangle$. By the Riesz theorem,*[9] *there exists a unique vector, which we denote by $T^\dagger|y\rangle$, such that*

$$F_y(|x\rangle) = \langle T^\dagger y|x\rangle, \qquad \forall\, |x\rangle \in \mathcal{H}. \tag{125}$$

*This defines the action of $T^\dagger$ on each vector $|y\rangle$. Uniqueness in Riesz shows that $T^\dagger$ is linear: the vector representing $F_{\alpha y_1+\beta y_2}$ is $\alpha T^\dagger|y_1\rangle + \beta T^\dagger|y_2\rangle$. Moreover, the previous estimate gives*

$$\|T^\dagger|y\rangle\| \leq M_T\,\||y\rangle\|, \tag{126}$$

*so $T^\dagger$ is also a bounded operator.*

*In finite dimension, this construction recovers the usual matrix operation. More precisely, if $\mathcal{H}$ is finite-dimensional and we choose an orthonormal basis, then, if $A$ is the matrix of $T$ in that basis, the matrix of $T^\dagger$ is*

$$A^\dagger = \overline{A}^T, \tag{127}$$

*that is, the conjugate transpose.* □

## 3.2 Hermitian Conjugation Identities in Dirac Notation

We now carefully derive the identities that relate the adjoint to bras and kets. These identities are the working rules that make the notation $C^\dagger$ usable in practice.

### 3.2.1 Two equivalent forms of the defining identity

Definition (121) says directly that

$$\boxed{\langle y|T|x\rangle = \langle T^\dagger y|x\rangle.} \tag{128}$$

This form is very useful in Dirac notation: it allows one to move the operator $T$ from the ket $|x\rangle$ to the bra $\langle y|$, replacing it by its adjoint $T^\dagger$.

Taking complex conjugates on both sides of (128), we recover the equivalent form

$$\overline{\langle y|T|x\rangle} = \langle x|T^\dagger|y\rangle. \tag{129}$$

To obtain this equality we use the elementary identity of the inner product

$$\overline{\langle a|b\rangle} = \langle b|a\rangle, \tag{130}$$

Here the expression $\langle x|T^\dagger|y\rangle$ must be read associatively as

$$\langle x|T^\dagger|y\rangle := \langle x|\big(T^\dagger|y\rangle\big). \tag{131}$$

That is, in this writing the operator $T^\dagger$ first acts on the ket $|y\rangle$, and then the bra $\langle x|$ evaluates the resulting vector. If we want the operator to act on the bra, we must write this explicitly as $\langle x|T^\dagger$.

---

[9] Frigyes Riesz (1880–1956) was a Hungarian mathematician and one of the central figures in the birth of modern functional analysis. The Riesz representation theorem for Hilbert spaces states that every bounded linear functional $F : \mathcal{H} \to \mathbb{C}$ can be written uniquely as

$$F(|x\rangle) = \langle z|x\rangle \tag{XI}$$

for a unique vector $|z\rangle \in \mathcal{H}$. Moreover, $\|F\| = \||z\rangle\|$. This result is precisely the bridge that allows one to convert bounded linear functionals into vectors of the Hilbert space.

### 3.2.2 The operator–bra identity

Since $T|x\rangle = |Tx\rangle$, one may ask how the ket $T|x\rangle$ transforms under Hermitian conjugation. The correct answer is an identity between bras:

$$\boxed{\big(T|x\rangle\big)^\dagger = \langle x|T^\dagger.} \tag{132}$$

To see this without ambiguity, let us evaluate both sides on an arbitrary ket $|y\rangle$. On one hand,

$$\big((T|x\rangle)^\dagger\big)(|y\rangle) = \langle Tx, y\rangle = \overline{\langle y|T|x\rangle}. \tag{133}$$

By the definition of the adjoint, the last term is

$$\overline{\langle y|T|x\rangle} = \langle x|T^\dagger|y\rangle. \tag{134}$$

On the other hand, $\langle x|T^\dagger$ is the new bra defined by precomposition with $T^\dagger$, that is,

$$\big(\langle x|T^\dagger\big)(|y\rangle) := \langle x|\big(T^\dagger|y\rangle\big) = \langle x|T^\dagger|y\rangle. \tag{135}$$

Since both bras give the same scalar for every $|y\rangle$, they are the same linear functional. Thus there is no ambiguity: in $T^\dagger|y\rangle$ the operator acts to the right on the ket; in $\langle x|T^\dagger$ the operator composes on the right with the bra and produces a new bra.

This identity is the operational content of the rule that physicists often state informally as: *taking the Hermitian conjugate moves an operator from the ket to the bra, replacing it by its adjoint.*

**Remark 3** (On a common notational abbreviation)**.** *The expression $\langle x\,T^\dagger|$ is sometimes seen informally as an abbreviation for $\langle x|T^\dagger$, since it arises directly from $(T|x\rangle)^\dagger$. Strictly speaking, however, $\langle x\,T^\dagger|$ is not standard Dirac notation; when an author writes it, what is meant is precisely $\langle x|T^\dagger$, that is, the bra $\langle x|$ on which the operator $T^\dagger$ acts from the right.*

### 3.2.3 Summary of the fundamental identities

With the convention above, one must distinguish two uses of conjugation: if the expression is a complex scalar, we use the bar $^-$ for complex conjugation; if the expression involves operators, kets, or bras, we use † for the Hermitian adjoint. The fundamental identities are:

$$\langle y|T|x\rangle = \langle T^\dagger y|x\rangle, \tag{136}$$

$$\overline{\langle y|T|x\rangle} = \langle x|T^\dagger|y\rangle, \tag{137}$$

$$\big(T|x\rangle\big)^\dagger = \langle x|T^\dagger. \tag{138}$$

Equation (136) is the usual form of the definition of the adjoint; (137) is the same identity after conjugating the scalar $\langle y|T|x\rangle$; and (138) describes how a ket transformed by $T$ becomes a bra on which $T^\dagger$ acts from the right. The last identity is often the most intuitive way to remember the operational rule: taking the adjoint moves the operator from the ket to the bra and replaces it by its adjoint.

**Example 9** (Unitary matrices $SU(2)$ and Pauli matrices)**.** *In quantum mechanics one normally writes $A^\dagger$ instead of $A^*$. A* qubit *is a two-level quantum system whose state space is modeled by $\mathbb{C}^2$. Thus, the algebra $M_2(\mathbb{C})$ of matrix operators on a qubit can be viewed as a $C^\dagger$-algebra, with*

$$A^\dagger := \overline{A}^{\,T}. \tag{139}$$

*Inside this algebra, a special role is played by the group*

$$SU(2) := \{U \in M_2(\mathbb{C}) : U^\dagger U = I,\ \det(U) = 1\}. \tag{140}$$

*Its elements are unitary matrices of determinant one. Here* det *denotes the determinant of a matrix. In physics, these matrices model reversible transformations of a qubit that preserve the norm of the state. To describe* $2 \times 2$ *matrices concretely, one introduces the* Pauli matrices*:*

$$\sigma_x := \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y := \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_z := \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{141}$$

*With the physical notation for the adjoint, the Pauli matrices satisfy*

$$\sigma_x^\dagger = \sigma_x, \qquad \sigma_y^\dagger = \sigma_y, \qquad \sigma_z^\dagger = \sigma_z. \tag{142}$$

*That is, they are Hermitian. Moreover,*

$$\mathrm{Tr}(\sigma_x) = \mathrm{Tr}(\sigma_y) = \mathrm{Tr}(\sigma_z) = 0, \tag{143}$$

*so* $\sigma_x, \sigma_y, \sigma_z$ *form a real basis of the subspace of traceless Hermitian* $2\times 2$ *matrices. The trace* $\mathrm{Tr}(A)$ *of a square matrix* $A = (a_{ij})$ *is the sum of its diagonal entries:*

$$\mathrm{Tr}(A) := a_{11} + a_{22} + \cdots + a_{nn}. \tag{144}$$

*We now pass to the infinitesimal aspect. The* Lie algebra *of a matrix group is, informally, the space of tangent vectors to the group at the identity; it encodes the infinitesimal transformations which, when exponentiated, produce elements of the group. To see what the Lie algebra of* $SU(2)$ *must be, take a differentiable curve*

$$U : (-\varepsilon, \varepsilon) \longrightarrow SU(2), \qquad U(0) = I, \tag{145}$$

*and call*

$$X := U'(0) \tag{146}$$

*its tangent vector at the identity. Since* $U(t)^\dagger U(t) = I$ *for all* $t$*, differentiating at* $t = 0$ *gives*

$$X^\dagger + X = 0. \tag{147}$$

*Therefore* $X$ *is anti-Hermitian. Let us write* $\delta x$ *for an infinitesimal variation of the curve parameter around* $0$*. By Taylor expansion, entry by entry,*

$$U(\delta x) = U(0) + \delta x\, U'(0) + O((\delta x)^2). \tag{148}$$

*Since* $U(0) = I$ *and* $U'(0) = X$*, we have*

$$U(\delta x) = I + \delta x\, X + O((\delta x)^2), \tag{149}$$

*where* $O((\delta x)^2)$ *denotes terms of order at least two in the infinitesimal variation* $\delta x$*. Applying the determinant, we use its first-order expansion at the identity:*

$$1 = \det(U(\delta x)) = 1 + \delta x\ \mathrm{Tr}(X) + O((\delta x)^2). \tag{150}$$

*Subtracting* $1$ *on both sides gives*

$$0 = \delta x\ \mathrm{Tr}(X) + O((\delta x)^2). \tag{151}$$

*Since this holds for any infinitesimal variation $\delta x \neq 0$, the first-order coefficient must vanish; therefore*

$$\mathrm{Tr}(X) = 0. \tag{152}$$

*Thus, every matrix tangent to $SU(2)$ at the identity is anti-Hermitian and traceless.*

*Conversely, suppose that $X^\dagger = -X$ and $\mathrm{Tr}(X) = 0$. Then*

$$U(t) := e^{tX} \tag{153}$$

*satisfies*

$$U(t)^\dagger U(t) = I, \qquad \det(U(t)) = e^{t\,\mathrm{Tr}(X)} = 1. \tag{154}$$

*Therefore $U(t) \in SU(2)$. Thus the Lie algebra of $SU(2)$ is*

$$T_I SU(2) = \mathfrak{su}(2) := \{X \in M_2(\mathbb{C}) : X^\dagger = -X,\ \mathrm{Tr}(X) = 0\}. \tag{155}$$

*The Pauli matrices do not belong directly to this tangent space, because they are Hermitian. To obtain tangent vectors one must multiply them by $-i/2$:*

$$\boxed{\begin{gathered} T_I SU(2) = \mathfrak{su}(2) \\ = \mathrm{span}_{\mathbb{R}} \left\{ -\frac{i}{2}\sigma_x,\ -\frac{i}{2}\sigma_y,\ -\frac{i}{2}\sigma_z \right\}. \end{gathered}} \tag{156}$$

*These three elements are anti-Hermitian, have trace zero, and form a real basis of $\mathfrak{su}(2)$. Equivalently, in the physics convention one writes the Hermitian operators*

$$J_x := \frac{\sigma_x}{2}, \qquad J_y := \frac{\sigma_y}{2}, \qquad J_z := \frac{\sigma_z}{2}, \tag{157}$$

*which satisfy the commutation relations*

$$[J_x, J_y] = iJ_z, \qquad [J_y, J_z] = iJ_x, \qquad [J_z, J_x] = iJ_y. \tag{158}$$

*Here the commutator of two operators is defined by*

$$[A, B] := AB - BA. \tag{159}$$

*Exponentiating gives, for each axis $j \in \{x, y, z\}$, a family of matrices in $SU(2)$:*

$$U_j(\theta) := \exp(-i\theta J_j) = \exp\!\left(-\frac{i\theta}{2}\sigma_j\right). \tag{160}$$

*If $\delta\theta$ denotes an infinitesimal angle, then the Taylor expansion of the exponential gives*

$$U_j(\delta\theta) = \exp(-i\,\delta\theta\,J_j) = \exp\!\left(-\frac{i\,\delta\theta}{2}\sigma_j\right) = I - \frac{i\,\delta\theta}{2}\sigma_j + O((\delta\theta)^2). \tag{161}$$

*To see geometrically what these unitary matrices do to a qubit, we must first understand the space of pure physical states.* A pure state *is a state described by a single normalized vector of the Hilbert space, up to global phase. In a two-level system, a normalized state is written as*

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \qquad |\alpha|^2 + |\beta|^2 = 1, \tag{162}$$

*with $\alpha, \beta \in \mathbb{C}$. Since $\alpha$ and $\beta$ contain four real parameters and the normalization condition imposes one real constraint, normalized vectors form a three-dimensional sphere $S^3 \subset \mathbb{R}^4$. However, in quantum mechanics the vectors*

$$|\psi\rangle \qquad y \qquad e^{i\chi}|\psi\rangle \tag{163}$$

*represent the same physical state, because they differ only by a global phase. Therefore, the pure physical states of a qubit are not the points of $S^3$, but the equivalence classes*

$$S^3/U(1) \cong S^2. \tag{164}$$

*Here $U(1) := \{e^{i\chi} : \chi \in \mathbb{R}\}$ is the group of complex phases of modulus 1, and quotienting by $U(1)$ means identifying vectors that differ only by a global phase. This two-dimensional sphere is the* Bloch sphere.[10]

*A concrete way to put coordinates on that sphere is to use the Pauli matrices. We define the Bloch vector associated with $|\psi\rangle$ by*

$$r_\psi := \big(\langle\psi|\sigma_x|\psi\rangle,\, \langle\psi|\sigma_y|\psi\rangle,\, \langle\psi|\sigma_z|\psi\rangle\big) \in \mathbb{R}^3. \tag{165}$$

*Since the Pauli matrices are Hermitian, these three quantities are real. If $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, then, in the computational basis,*

$$|\psi\rangle = \begin{pmatrix}\alpha\\ \beta\end{pmatrix}, \qquad \langle\psi| = \begin{pmatrix}\overline{\alpha} & \overline{\beta}\end{pmatrix}. \tag{166}$$

*First we compute the action of each Pauli matrix on the ket:*

$$\sigma_x|\psi\rangle = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\begin{pmatrix}\alpha\\ \beta\end{pmatrix} = \begin{pmatrix}\beta\\ \alpha\end{pmatrix}, \tag{167}$$

$$\sigma_y|\psi\rangle = \begin{pmatrix}0 & -i\\ i & 0\end{pmatrix}\begin{pmatrix}\alpha\\ \beta\end{pmatrix} = \begin{pmatrix}-i\beta\\ i\alpha\end{pmatrix}, \tag{168}$$

$$\sigma_z|\psi\rangle = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\begin{pmatrix}\alpha\\ \beta\end{pmatrix} = \begin{pmatrix}\alpha\\ -\beta\end{pmatrix}. \tag{169}$$

*Now we take the inner product with $\langle\psi|$:*

$$\langle\psi|\sigma_x|\psi\rangle = \begin{pmatrix}\overline{\alpha} & \overline{\beta}\end{pmatrix}\begin{pmatrix}\beta\\ \alpha\end{pmatrix} = \overline{\alpha}\beta + \overline{\beta}\alpha = 2\,\mathrm{Re}(\overline{\alpha}\beta), \tag{170}$$

$$\langle\psi|\sigma_y|\psi\rangle = \begin{pmatrix}\overline{\alpha} & \overline{\beta}\end{pmatrix}\begin{pmatrix}-i\beta\\ i\alpha\end{pmatrix} = -i\,\overline{\alpha}\beta + i\,\overline{\beta}\alpha = 2\,\mathrm{Im}(\overline{\alpha}\beta), \tag{171}$$

$$\langle\psi|\sigma_z|\psi\rangle = \begin{pmatrix}\overline{\alpha} & \overline{\beta}\end{pmatrix}\begin{pmatrix}\alpha\\ -\beta\end{pmatrix} = |\alpha|^2 - |\beta|^2. \tag{172}$$

*Therefore,*

$$r_\psi = \big(2\,\mathrm{Re}(\overline{\alpha}\beta),\, 2\,\mathrm{Im}(\overline{\alpha}\beta),\, |\alpha|^2 - |\beta|^2\big). \tag{173}$$

[10]The name recalls Felix Bloch (1905–1983). In his classic work on nuclear induction, Bloch introduced a vectorial description of spin dynamics through magnetization; see F. Bloch, "Nuclear Induction", *Physical Review* **70** (1946), 460–474. In quantum information, the geometric representation of the pure state space of a two-level system is known precisely as the Bloch sphere.

*The reason the complex number $\overline{\alpha}\beta$ appears is that the first two coordinates of the Bloch vector are precisely its real part and its imaginary part, multiplied by 2. If we write $z := \overline{\alpha}\beta$, then*

$$\|r_\psi\|^2 = 4\big(\operatorname{Re} z\big)^2 + 4\big(\operatorname{Im} z\big)^2 + \big(|\alpha|^2 - |\beta|^2\big)^2 \tag{174}$$

$$= 4|z|^2 + \big(|\alpha|^2 - |\beta|^2\big)^2 \tag{175}$$

$$= 4|\alpha|^2|\beta|^2 + \big(|\alpha|^2 - |\beta|^2\big)^2 \tag{176}$$

$$= \big(|\alpha|^2 + |\beta|^2\big)^2 = 1. \tag{177}$$

*In summary, we start with a normalized ket $|\psi\rangle \in \mathbb{C}^2$ and, by taking the three expectation values of the Pauli matrices, we produce a real vector $r_\psi \in \mathbb{R}^3$. The previous calculation shows that this vector has norm 1: it is not an arbitrary vector of $\mathbb{R}^3$, but a point on a sphere. Here $z = \overline{\alpha}\beta$ was only an auxiliary complex number used to abbreviate the first two coordinates; it is not $z$ that lives on the Bloch sphere. The original normalized vector $(\alpha, \beta) \in \mathbb{C}^2$ lives in $S^3 \subset \mathbb{R}^4$, whereas the Bloch vector $r_\psi$ lives on the sphere*

$$\boxed{S^2 := \{r \in \mathbb{R}^3 : \|r\| = 1\}}, \tag{178}$$

*which is precisely the Bloch sphere.*

*Now let us see why the unitary matrices in $SU(2)$ produce rotations of that sphere. If $U \in SU(2)$, the coordinates of the transformed state $U|\psi\rangle$ are computed as follows. Recall that the bra corresponding to this new ket is obtained by taking the adjoint: $\langle U\psi| := (U|\psi\rangle)^\dagger = \langle\psi|U^\dagger$. Therefore,*

$$\big(r_{U\psi}\big)_j = \langle U\psi|\sigma_j|U\psi\rangle = \langle\psi|U^\dagger\sigma_j U|\psi\rangle. \tag{179}$$

*Let us do the explicit calculation for the rotation around the $z$-axis. In this case*

$$U_z(\theta) = \exp\Big(-\frac{i\theta}{2}\sigma_z\Big) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}. \tag{180}$$

*Before substituting coordinates, let us compute directly the matrices that appear in*

$$\langle\psi|U_z(\theta)^\dagger\sigma_j U_z(\theta)|\psi\rangle. \tag{181}$$

*Here $r_\psi = (r_x, r_y, r_z)$ means*

$$r_x := \langle\psi|\sigma_x|\psi\rangle, \qquad r_y := \langle\psi|\sigma_y|\psi\rangle, \qquad r_z := \langle\psi|\sigma_z|\psi\rangle. \tag{182}$$

*For $\sigma_x$,*

$$U_z(\theta)^\dagger\sigma_x U_z(\theta) = \begin{pmatrix} e^{i\theta/2} & 0 \\ 0 & e^{-i\theta/2} \end{pmatrix}\begin{pmatrix} 0 & e^{i\theta/2} \\ e^{-i\theta/2} & 0 \end{pmatrix} \tag{183}$$

$$= \begin{pmatrix} 0 & e^{i\theta} \\ e^{-i\theta} & 0 \end{pmatrix} = (\cos\theta)\sigma_x - (\sin\theta)\sigma_y. \tag{184}$$

*For $\sigma_y$,*

$$U_z(\theta)^\dagger\sigma_y U_z(\theta) = \begin{pmatrix} e^{i\theta/2} & 0 \\ 0 & e^{-i\theta/2} \end{pmatrix}\begin{pmatrix} 0 & -ie^{i\theta/2} \\ ie^{-i\theta/2} & 0 \end{pmatrix} \tag{185}$$

$$= \begin{pmatrix} 0 & -ie^{i\theta} \\ ie^{-i\theta} & 0 \end{pmatrix} = (\sin\theta)\sigma_x + (\cos\theta)\sigma_y. \tag{186}$$

*And for $\sigma_z$,*

$$U_z(\theta)^\dagger \sigma_z U_z(\theta) = \begin{pmatrix} e^{i\theta/2} & 0 \\ 0 & e^{-i\theta/2} \end{pmatrix} \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & -e^{i\theta/2} \end{pmatrix} \tag{187}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \sigma_z. \tag{188}$$

*By definition, the $j$-th coordinate of the Bloch vector associated with the transformed state $U_z(\theta)|\psi\rangle$ is*

$$\left(r_{U_z(\theta)\psi}\right)_j := \langle U_z(\theta)\psi|\sigma_j|U_z(\theta)\psi\rangle = \langle \psi|U_z(\theta)^\dagger \sigma_j U_z(\theta)|\psi\rangle. \tag{189}$$

*By sandwiching the three previous identities with $\langle\psi|$ and $|\psi\rangle$, we obtain the coordinates of the new Bloch vector:*

$$\left(r_{U_z(\theta)\psi}\right)_x = \langle \psi|U_z(\theta)^\dagger \sigma_x U_z(\theta)|\psi\rangle \tag{190}$$

$$= \cos\theta\,\langle\psi|\sigma_x|\psi\rangle - \sin\theta\,\langle\psi|\sigma_y|\psi\rangle = r_x\cos\theta - r_y\sin\theta, \tag{191}$$

$$\left(r_{U_z(\theta)\psi}\right)_y = \langle \psi|U_z(\theta)^\dagger \sigma_y U_z(\theta)|\psi\rangle \tag{192}$$

$$= \sin\theta\,\langle\psi|\sigma_x|\psi\rangle + \cos\theta\,\langle\psi|\sigma_y|\psi\rangle = r_x\sin\theta + r_y\cos\theta, \tag{193}$$

$$\left(r_{U_z(\theta)\psi}\right)_z = \langle \psi|U_z(\theta)^\dagger \sigma_z U_z(\theta)|\psi\rangle \tag{194}$$

$$= \langle\psi|\sigma_z|\psi\rangle = r_z. \tag{195}$$

*Therefore,*

$$r_{U_z(\theta)\psi} = \left(r_x\cos\theta - r_y\sin\theta,\, r_x\sin\theta + r_y\cos\theta,\, r_z\right). \tag{196}$$

*That is,*

$$r_{U_z(\theta)\psi} = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} r_\psi. \tag{197}$$

*This is exactly the ordinary rotation matrix around the $z$-axis on the Bloch sphere. If the same calculation is repeated with $U_x(\theta)$ and $U_y(\theta)$, one obtains the ordinary rotations around the $x$ and $y$ axes, respectively. We denote by $SO(3)$ the group of real rotations of $\mathbb{R}^3$, that is,*

$$SO(3) := \{R \in M_3(\mathbb{R}) : R^T R = I,\ \det(R) = 1\}. \tag{198}$$

*The previous calculation explicitly shows how a unitary of the form $U_z(\theta)$ produces an ordinary rotation around the $z$-axis. Similarly, $U_x(\theta)$ and $U_y(\theta)$ produce rotations around the $x$ and $y$ axes. In general, to each unitary $U \in SU(2)$ there corresponds a real rotation $R_U \in SO(3)$ such that the action on Bloch vectors is written as*

$$\boxed{r_{U\psi} = R_U r_\psi, \qquad R_U \in SO(3).} \tag{199}$$

*A normalized pure state of a qubit determines a point on the Bloch sphere $S^2$. A unitary operator $U \in SU(2)$ transforms qubit states and, when passing to Bloch vectors, induces a rotation $R_U \in SO(3)$ of the Bloch sphere.*

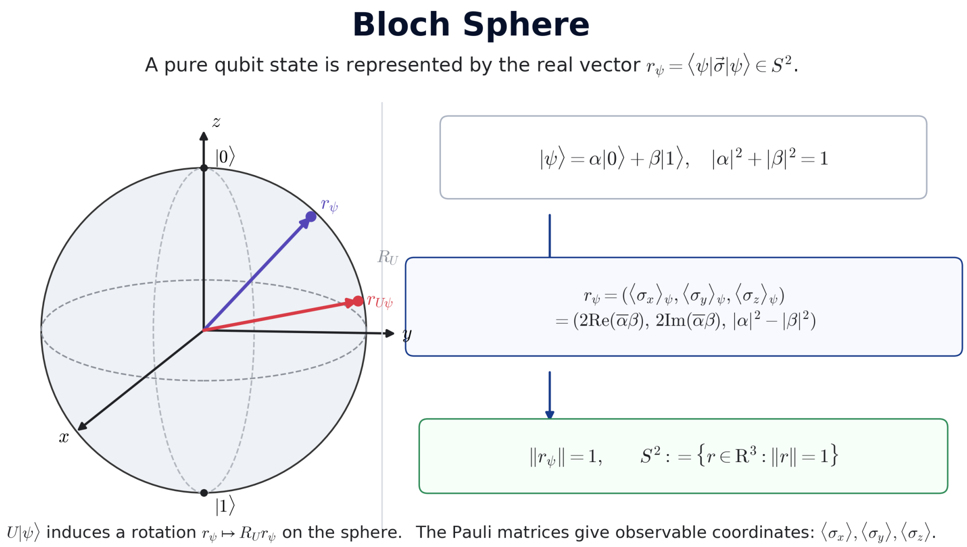


**Figure 3:** Bloch sphere: to a normalized ket $|\psi\rangle$ one associates the real vector $r_\psi = \langle\psi|\vec{\sigma}|\psi\rangle \in S^2$. The unitary action $U|\psi\rangle$ induces a rotation $r_\psi \mapsto R_U r_\psi$ on the sphere.

□

# 4 Notes on Unbounded Operators

In quantum mechanics, not all important operators are bounded. In fact, the most basic observables in infinite dimension are often unbounded operators. An *observable* is a measurable physical quantity, mathematically represented by a self-adjoint operator; the *Hamiltonian* is the energy observable that generates the time evolution of the system. Typical examples are position, momentum, and many Hamiltonians. For this reason it is useful to separate two worlds:

$$B(\mathcal{H}) \quad \text{contains bounded operators defined on all of } \mathcal{H}, \tag{200}$$

whereas unbounded operators are normally defined only on a dense part of the Hilbert space. The difference is not a minor technical subtlety: it changes what it means to compose operators, take adjoints, and write equations such as $T|\psi\rangle$.

**Definition 9** (Linear operator with domain)**.** *Let $\mathcal{H}$ be a Hilbert space. A linear operator, not necessarily bounded, is written as*

$$T : \mathrm{Dom}(T) \subseteq \mathcal{H} \longrightarrow \mathcal{H}, \tag{201}$$

*where* $\mathrm{Dom}(T)$ *is a vector subspace called the* domain *of $T$. Saying that $T$ is linear means that*

$$T(\alpha|\psi\rangle + \beta|\phi\rangle) = \alpha T|\psi\rangle + \beta T|\phi\rangle \tag{202}$$

*whenever $|\psi\rangle, |\phi\rangle \in \mathrm{Dom}(T)$ and $\alpha, \beta \in \mathbb{C}$.*

*The operator is called* bounded on its domain *if there exists $M \geq 0$ such that*

$$\|T|\psi\rangle\| \leq M\||\psi\rangle\|, \qquad |\psi\rangle \in \mathrm{Dom}(T). \tag{203}$$

*If no such constant exists, we say that $T$ is* unbounded*. When a bounded operator is defined on all of $\mathcal{H}$, it belongs to $B(\mathcal{H})$.*

Take $\mathcal{H} = L^2(\mathbb{R})$. Here $L^2(\mathbb{R})$ denotes the space of complex square-integrable functions on $\mathbb{R}$, that is, functions $\psi$ such that

$$\int_{\mathbb{R}} |\psi(x)|^2 \, dx < \infty. \tag{204}$$

The inner product of $L^2(\mathbb{R})$ is

$$\langle f, g\rangle_{L^2} := \int_{\mathbb{R}} \overline{f(x)} g(x) \, dx. \tag{205}$$

With our definition of bra, this is written in Dirac notation as

$$\boxed{\langle \phi|\psi\rangle := \langle \phi, \psi\rangle_{L^2} = \int_{\mathbb{R}} \overline{\phi(x)} \psi(x) \, dx.} \tag{206}$$

**Example 10** (The position operator)**.** *The position operator is formally defined by*

$$(Q\psi)(x) := x\psi(x). \tag{207}$$

*But not every function $\psi \in L^2(\mathbb{R})$ satisfies $x\psi(x) \in L^2(\mathbb{R})$. Therefore the natural domain is*

$$\mathrm{Dom}(Q) := \{\psi \in L^2(\mathbb{R}) : x\psi(x) \in L^2(\mathbb{R})\}. \tag{208}$$

*This operator is unbounded. To see this, let $\psi_n$ be a function of norm 1 whose support is contained in the interval $[n, n+1]$. On that interval one has $|x| \geq n$, and outside that interval $\psi_n(x) = 0$. Therefore,*

$$\|Q\psi_n\|^2 = \int_{\mathbb{R}} |x|^2 |\psi_n(x)|^2 \, dx = \int_n^{n+1} |x|^2 |\psi_n(x)|^2 \, dx \geq n^2 \int_n^{n+1} |\psi_n(x)|^2 \, dx = n^2, \tag{209}$$

*and therefore $\|Q\psi_n\| \geq n$. Since $n$ can grow without bound, there is no constant $M$ satisfying $\|Q\psi\| \leq M\|\psi\|$ for every $\psi \in \mathrm{Dom}(Q)$.*

For unbounded operators one must always keep track of the domain. For example, an expression such as

$$ST|\psi\rangle \tag{210}$$

only makes sense if $|\psi\rangle \in \mathrm{Dom}(T)$ and, moreover, $T|\psi\rangle \in \mathrm{Dom}(S)$. Thus, the domain of the composition is

$$\mathrm{Dom}(ST) := \{|\psi\rangle \in \mathrm{Dom}(T) : T|\psi\rangle \in \mathrm{Dom}(S)\}. \tag{211}$$

This is one of the reasons why the formalism of unbounded operators is more delicate than that of matrices or bounded operators.

**Definition 10** (Adjoint of a densely defined operator)**.** *Let $T : \mathrm{Dom}(T) \subseteq \mathcal{H} \to \mathcal{H}$ be a linear operator whose domain is dense in $\mathcal{H}$. Saying that $\mathrm{Dom}(T)$ is dense means that every vector of $\mathcal{H}$ can be approximated arbitrarily well by vectors in $\mathrm{Dom}(T)$. The adjoint $T^\dagger$ is defined as follows: a vector $|y\rangle \in \mathcal{H}$ belongs to $\mathrm{Dom}(T^\dagger)$ if there exists a vector $|z\rangle \in \mathcal{H}$ such that*

$$\langle y|T|x\rangle = \langle z|x\rangle, \qquad |x\rangle \in \mathrm{Dom}(T). \tag{212}$$

*In that case we define*

$$T^\dagger|y\rangle := |z\rangle. \tag{213}$$

*The density of $\mathrm{Dom}(T)$ guarantees the uniqueness of $|z\rangle$: if two vectors produce the same functional on a dense subset, then they are equal.*

**Example 11** (The momentum operator and its adjoint)**.** *In the representation $\mathcal{H} = L^2(\mathbb{R})$, the momentum operator is formally written as*

$$P\psi := -i\hbar\,\frac{d\psi}{dx}. \tag{214}$$

*Here $\hbar$ is the reduced Planck constant. This operator is also not defined on all of $L^2(\mathbb{R})$: for $P\psi$ to make sense as a vector of $L^2(\mathbb{R})$, we need the derivative of $\psi$ to exist in the appropriate sense and to be square-integrable again. A usual domain is the Sobolev space*

$$\mathrm{Dom}(P) := H^1(\mathbb{R}), \tag{215}$$

*where $H^1(\mathbb{R})$ denotes the set of functions in $L^2(\mathbb{R})$ whose weak derivative*[11] *also belongs to $L^2(\mathbb{R})$.*

*The physical intuition is clear: differentiating measures oscillation. A family of increasingly oscillatory waves may have fixed $L^2$ norm, but increasingly large derivative. That is why momentum is an unbounded operator.*

*The choice of domain also contains the condition that allows integration by parts without producing boundary terms. On the real line, working with functions in $H^1(\mathbb{R})$ gives the natural framework for those identities to make sense; if we were working on a finite interval, we would have to impose boundary conditions explicitly, for example vanishing at the endpoints or periodic conditions.*

*With this domain, the adjoint of $P$ has the same differential expression. The formal reason is seen by comparing with the defining identity of the adjoint:*

$$\overline{\langle y|T|x\rangle} = \langle x|T^\dagger|y\rangle. \tag{216}$$

*Indeed, if $\phi$ and $\psi$ are sufficiently regular functions and vanish fast enough at the boundary, then*

$$\begin{aligned}
\overline{\langle\phi|P\psi\rangle} &= \overline{\int_{\mathbb{R}} \overline{\phi(x)}(-i\hbar\,\psi'(x))\,dx} && (217)\\
&= \int_{\mathbb{R}} \phi(x) i\hbar\,\overline{\psi'(x)}\,dx && (218)\\
&= \Big[i\hbar\,\phi(x)\overline{\psi(x)}\Big]_{-\infty}^{\infty} - \int_{\mathbb{R}} i\hbar\,\phi'(x)\overline{\psi(x)}\,dx && (219)\\
&= \int_{\mathbb{R}} \overline{\psi(x)}(-i\hbar\,\phi'(x))\,dx && (220)\\
&= \langle\psi|P\phi\rangle. && (221)
\end{aligned}$$

*Thus the defining identity holds:*

$$\overline{\langle\phi|P|\psi\rangle} = \langle\psi|P|\phi\rangle, \tag{222}$$

*and therefore $P$ is self-adjoint:*

$$\boxed{P^\dagger = P, \qquad \mathrm{Dom}(P^\dagger) = H^1(\mathbb{R}).} \tag{223}$$

□

[11]The weak derivative is defined by duality with test functions. We say that $g$ is the weak derivative of $f$ if, for every smooth compactly supported function $\varphi \in C_c^\infty(\mathbb{R})$, one has

$$\int_{\mathbb{R}} f(x)\varphi'(x)\,dx = -\int_{\mathbb{R}} g(x)\varphi(x)\,dx. \tag{XII}$$

If $f$ is differentiable in the usual sense, then $g = f'$. The advantage is that this definition still makes sense for less regular functions.

# 5 Conclusion

The notion of a $C^\dagger$-algebra, as it is sometimes found in the physics literature, is not an independent mathematical structure: it is definitionally identical to a $C^*$-algebra, with the abstract involution $*$ renamed as $\dagger$ to agree with the Hilbert adjoint notation used in quantum mechanics. This identification becomes precise once the algebra is concretely represented as bounded operators on a Hilbert space, at which point $a^* = a^\dagger$ for every element $a$. On the basis of this identification, the operator-theoretic identities (136)–(138) provide the standard working rules for manipulating adjoints, bras, and kets in Dirac notation.